

that allow a low-res model to be smoothed automatically at render time so that artists can get the best of both worlds.

Polygonal components

Polygonal surfaces have four major components: vertices, edges, faces, and elements. Editable mesh objects also have a triangular face sub-object, while editable poly objects have a border sub-object. To select one of these components, highlight an icon in the Selection rollout; select the sub-object in the modifier stack, or right-click over the surface to open a quad menu, where the sub-object can be selected. The types of polygonal components are:

- A vertex represents a single point in space; vertices are the building blocks of all polygonal objects.
- An edge connects two vertices, forming a line. Edges also connect the sides of polygons or faces.
- A border, in an editable poly object, is a connected loop of edges on the open border of an object.
- A face, in an editable mesh, is a triangular plane.
- A polygon, in a poly object, is a face that defines a plane; in an editable mesh, a polygon is a face composed of multiple triangles.
- An element is a polygon that is not attached to a polygonal object.

Polygonal objects can be edited by using either the Edit Mesh or Edit Poly modifier. Objects can also be collapsed to editable mesh or editable poly objects. The big difference with editable mesh and editable poly is that poly objects can have polygons with any number of sides, whereas mesh objects are based in triangles. The Edit Poly modifier also has additional selection and editing tools, plus the ability to perform paint deformations and create subdivision surfaces. The editable mesh object is the simplest of the polygonal object types, and is used when precise control over the polygons is needed, such as when creating low-poly models for games. For most applications, editable poly is the preferred geometry because it imposes fewer limitations on the modeler and has a more robust set of tools.

Edit Poly Modifier

The Edit Poly modifier converts an object into an editable poly object. This modifier allows you to reshape the object and to add or subtract detail. In addition, the modifier has controls that aid in texturing an object as well as paint deformation and animation tools.